WASP-12b

R-1903

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_160107` · created 2026-08-05T23:01:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457394.6909 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.042
Transit depth (Rp/Rs)^2	1.45 +/- 1.01 [%]
Orbital Inclination (inc)	77.06 +/- 2.89
Airmass coefficient 1 (a1)	0.977 +/- 0.034
Airmass coefficient 2 (a2)	0.021 +/- 0.024
Scatter in the residuals of the lightcurve fit is	3.51 %
Best Comparison Star	#2 - [63, 22]
Optimal Aperture	3.89
Optimal Annulus	5.19
Transit Duration (day)	0.103 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.121 ± 0.042 vs 0.1178 (deviation 0.06σ)
Duration fit vs published	0.103 d vs 0.125 d (deviation 1.04σ)
Mid-time O–C	-0.7 min
Depth SNR	2.3
Residual scatter	3.51 %

Light curve

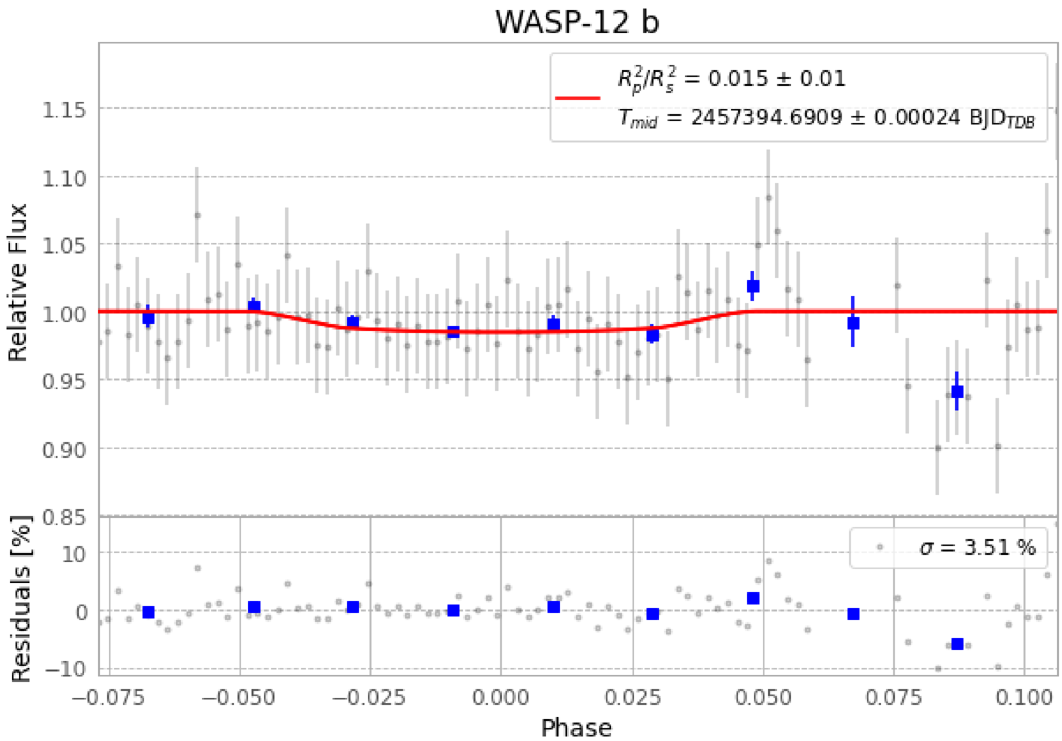

Figure 1 — FinalLightCurve_WASP-12 b_2016-01-06.png (SHA-256 c34874772066ff10…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 172], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 4ee2f7e16d77d197…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

97 FITS frames (64 MB), `WASP-12160107022412.FITS` → `WASP-12160107071212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160107.log (8127b8605abc…), FinalLightCurve_WASP-12 b_2016-01-06.png (c34874772066…), FinalLightCurve_WASP-12 b_2016-01-06.csv (86e7c8c620f6…)

Compute resources

Wall time 740.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)